<u>Point-by-point Response to Comments from five Reviewers</u>

First of all, we would like to express our sincere appreciation for five reviewers' in-depth critiques, which significantly helped us in preparing a revision of the paper. In the revised manuscript, we have tried our best to ***address all concerns from all reviewers***.

[Reviewer #1]

1. *The authors applied a semantics-oriented method to study microRNA regulation on glucocorticoid resistance in pediatric acute lymphoblastic leukemia (ALL). A semantic search tool, OmniSearch, was introduced to integrate multiple databases from different sources. That would be helpful for authors to show the difference between their new methods and traditional bioinformatics analysis about microRNA expression and regulation.*

   Response: No edits are needed.

[Reviewer #2]

1. *The claim that OMIT organizes MeSH in a BFO compliant manner is patentliy false. A MeSH term is not a quality. Numerous plural names. "Diseases Category" is a use-mention confusion and awkward English usage as well.*

   Response: First, quality is not a requirement for a term to follow the BFO backbone. Second, all MeSH terms, "Diseases Category" for example, were obtained from the official NCBI website, and we believe that it is not a good practice to change such terms.

[Reviewer #3]

1. *The omnisearch website, http://omnisearch.soc.southalabama.edu/index.php/Software, cannot be visited as of now.*

   Response: The OmniSearch site was down during the Nate hurricane. It has been back to normal since then.

2. *Manual search of the literature already narrowed the miRNAs to a very small set of only 18 miRNAs. The contribution of the downstream computational methods is questionable.*

   Response: We discussed that, the total number of targets for these 18 miRNAs is 232; that is, more than two hundred miRNA::target pairs need to be further investigated, which would be, if not impossible, too time-consuming and labor-intensive.

3. *This paper, https://www.nature.com/leu/journal/v23/n4/full/leu2008370a.html, mentioned miR15b and several other miRNAs. Some of them were not identified in this study. Actually, miR15b was in the original set of 18 miRNAs, but was filtered out using MeSH terms. Not sure whether this is a desired result.*

   Response: The whole point of utilizing the MeSH term filtering is to make the search query results as precise as possible, i.e., to avoid irrelevant results. Hsa-miR-15b-3p was filtered because even with the Broader-Match option (the user-provided MeSH term and all of its parent terms), there were still no targets returned from the search. This indicated that there is no supporting evidence of the association between hsa-miR-15b-3p and leukemia, although hsa-miR-15b-3p is related to some other diseases.

[Reviewer #4]

1. *The authors declare that they disclose two miRNAs that are likely regulate GC resistance in pediatric ALL. This work is actually useful to assist with unraveling miRNA regulation mechanisms, but their results need to further confirm.*

   Response: We discussed that results reported in this paper were based on semantics-oriented computational analysis and reviewed by five domain experts. In addition, wet-lab validation will be performed in future work, which will further confirm our findings reported in this paper.

[Reviewer #5]

1. *The paper is well-written. The result is of importance in biomedical domain. The method used in this paper can be applied to other similar problems as well.*

   Response: No edits are needed.